

Hsin-Ling (Justin) Hsu

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Education	
National Chengchi University (NCCU) <i>B.S. Double Major in MIS and Computer Science</i> • GPA: 4.15 / 4.30 (3.96 / 4.00) Honors: Beta Gamma Sigma Honor Society (Top 10% of department). • Transferred from NCCU Department of Mathematical Sciences after freshman year.	Sep. 2023 – Expected Graduation: Jun. 2027 Taipei, Taiwan

Research Interests
Trustworthy VLMs/LLMs; AI / Computational Methods for Healthcare and Biomedicine.

Publications & On-Going Work
Total citations: 47; h-index: 2; i10-index: 2 (Google Scholar, May 2026)

Published / Accepted

- [1] **Hsin-Ling Hsu***, Cong-Tinh Dao*, et al., “MedPlan: A Two-Stage RAG-Based System for Personalized Medical Plan Generation.” *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025): Industry Track*, Vienna, Austria, 2025. (paper)
Notes: A* top conference, ~25.6% acceptance rate. 30+ citations.
- [2] Chia-Hsuan Hsu, Jun-En Ding, **Hsin-Ling Hsu**, et al., “RGPO: Ranking-Guided Preference Optimization for Reliable Clinical Reasoning.” *IEEE Transactions on Artificial Intelligence (IEEE TAI, Q1 Journal)*, 2026. (paper)
- [3] **Hsin-Ling Hsu**, et al., “KAP: MLLM-assisted OCR Text Enhancement for Hybrid Retrieval in Chinese Non-Narrative Documents.” *Proceedings of the FinTech in AI CUP Special Session, 18th NTCIR Conference*, Tokyo, Japan, 2025. (paper)
Notes: Oral Presentation. Travel grant from ROC Ministry of Education (\$3,600 USD). National Excellence Prize at AI CUP 2024.

Under Review

- [4] **Hsin-Ling Hsu**, et al., “MedAction: Towards Active Multi-turn Clinical Diagnostic LLMs.” *Submitted to COLM 2026*.
- [5] **Hsin-Ling Hsu**, et al., “WARP: Guaranteed Inner-Layer Repair of NLP Transformers.” *Submitted to COLM 2026*. (paper)
- [6] Jerry Wang, **Hsin-Ling Hsu**, et al., “Gradient-Free Adversarial Prompt Injection on RAG via Differential Evolution.” *Submitted to QRS 2026*.
- [7] **Hsin-Ling Hsu** and Jengnan Tzeng, “DAT: Dynamic Alpha Tuning for Hybrid Retrieval in Retrieval-Augmented Generation.” *arXiv*, 2025. (paper)
Notes: 10+ citations.

Patents
• ROC Utility Model Patent, No. 114209930 (First Inventor)

Research Experience	
Research Collaborator <i>BioMed-AI Lab, ECE/EECS, University of Michigan (Advisor: Prof. Liyue Shen)</i> • Research on AI for healthcare, focusing on multi-turn clinical diagnostic reasoning with LLMs, knowledge graphs, and chain-of-reasoning-action consistency. [4]	Feb. 2026 – Present Remote
Research Intern <i>FLAIR Lab, CSE, Texas A&M University (Advisor: Prof. Kuan-Hao Huang)</i> • Research on vision-language alignment and interpretability in VLMs, focusing on cross-modal representations via sparse autoencoders.	Nov. 2025 – Present Remote
Research Assistant <i>Far Eastern Memorial Hospital (Advisor: Dr. Fang-Ming Hung)</i> • Research on explainable clinical decision support systems, integrating VLMs/LLMs, reinforcement learning, and knowledge graphs for disease prediction and medical plan generation. [1][2][4]	Dec. 2024 – Present New Taipei, Taiwan
Research Assistant <i>Reinforcement Learning and Games Lab, Institute of Information Science, Academia Sinica (Advisor: Prof. Ti-Rong Wu)</i> • RL infrastructure research: game environments and AI solution tree visualization.	May 2024 – Present Taipei, Taiwan

Industry Experience

AI Intern

Sep. 2024 – Jun. 2025

GoFreight (The world's largest cloud-based freight forwarding software)

Taipei, Taiwan

- Leveraged MLLM parallelization for logistics OCR/NLP, reducing processing time by ~67% (45s to 15s).
- Built LLM-based web parsers with up to 97% accuracy.

AI Engineer (Part-Time)

Jul. 2023 – Sep. 2024

ChainSea Information Group

Taipei, Taiwan

- Core R&D on LLM/Whisper projects (transcription, inference acceleration, LoRA tuning).
- Built RAG pipelines and an LLM Agent for addiction counseling.

Selected Competitions & Honors

22nd National Innovation Award

2025

Project: MedPlan (ACL 2025). Co-participating with Far Eastern Memorial Hospital.

Taiwan

- First Inventor of patent. MOU signed with clinics/system vendors; deployed in clinical trials at hospital.

2nd Place, HOTAI MaaS Hackathon, [2/233 teams; ~0.8%]

2024

AI Travel Itinerary Health Check. Won NT\$250,000 prize. Tech: React.js, Hybrid RAG, Cross-Encoder.

Taiwan

3rd Place, LINE FRESH Campus Competition, [3/165 teams; ~1.8%]

2024

AI dementia care platform. Built backend, health tracking, and multilingual chatbot with LLMs.

Taiwan

CPE Collegiate Programming Examination — Professional Level (A), Top 4.6%

2025

Ranked 113/2481. ICPC-style algorithmic problem solving.

Taiwan

Academic Service & Other Activities

- **Reviewer / Program Committee:** KDD AI4Sci Track, NeurIPS Workshop on GenAI for Health, NeurIPS Workshop on Efficient Reasoning, Texas NLP Symposium, IEEE TCE, WMW.
- **Talks:** PyCon TW (academic research sharing); Google Developer Group at NCCU (Generative AI, Databases).
- **Volunteer:** Soobi rural children's programming camp instructor (Meowbit game design).
- **Columnist:** Customer Insights magazine (two columns on generative AI applications).
- **Open Source:** Core contributor to an LLM project with 300+ GitHub Stars and 30+ forks.
- **Security:** Two ZeroDay vulnerability disclosures.
- **Industry Projects:** Taipower smart AI chatbot ("Dianbao") training data optimization (during tenure at ChainSea).

Skills

Python, C/C++, PyTorch, Unsloth, veRL, Transformers, Flask, Langfuse, SQL, Linux, GCP, Docker, Git, HTML/CSS/JS